

Experiment Tracking

snowflakeR exposes **ML experiment tracking** on Snowflake (runs, parameters, metrics, artifacts, and logged models) through a small R API on top of the snowflake-ml-python SDK. Experiment tracking requires a recent snowflake-ml-python build (see ?sfr_experiment for the minimum version enforced in your session).

For **model versioning and deployment**, continue to use the Model Registry (vignette("model-registry")). Experiments complement the registry by capturing training exploration and tidymodels tune grids.

Basic workflow

```
library(snowflakeR)

conn <- sfr_connect()
exp <- sfr_experiment(conn, name = "MY_EXPERIMENT")

sfr_start_run(exp, name = "run-001")

sfr_exp_log_params(exp, alpha = 0.1, max_depth = 6L)
sfr_exp_log_metric(exp, "rmse", 0.42, step = 1L)

# Optional: log a model or artifact path
# sfr_exp_log_model(exp, ...)
# sfr_exp_log_artifact(exp, "/path/to/file", "plots/curve.png")

sfr_end_run(exp)
```

Listing and downloading artifacts

```
sfr_exp_list_artifacts(exp, run_name = "run-001")
sfr_exp_download_artifact(exp, run_name = "run-001",
                           artifact_path = "plots/curve.png",
                           target_path   = "curve.png")
```

Integration with tidymodels tune

Use sfr_experiment_from_tune() to attach an experiment to a tuning workflow, and sfr_experiment_log_best() to record the best result after tune::fit_resamples() or last_fit(). See the function help pages for the exact signatures expected by your tune version.

```
# Pseudocode -- adapt to your workflow and resampling object
# exp <- sfr_experiment(conn, "MY_EXPERIMENT")
# sfr_experiment_from_tune(exp, <tune_workflow_or_last_fit>)
# ... fit / tune ...
# sfr_experiment_log_best(exp, <results>)
```

Cleanup

```
sfr_delete_run(exp, run_name = "run-001")  
sfr_delete_experiment(exp, name = "MY_EXPERIMENT")
```

See also

- ?sfr_experiment, ?sfr_start_run, ?sfr_end_run
- ?sfr_exp_log_param, ?sfr_exp_log_metric, ?sfr_exp_log_model
- ?sfr_experiment_from_tune, ?sfr_experiment_log_best